

Seminar Notes on Kodaira's  
*Complex Manifolds and Deformation of Complex Structures,*  
*I*

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# Preface

These notes were written for a reading seminar on K. Kodaira, *Complex Manifolds and Deformation of Complex Structures, I* (1968). Their main aim is to fill in, by hand, the derivations and details that the book omits.

## Principles for these seminar notes

1. Prove what is written in the text; reconstruct each claim in one's own words.
2. Abbreviate notation and anything else clear from context as convenient.
3. Prioritize understanding: explain every point that is omitted or glossed over.
4. Reconstruct proofs oneself, or revise the flow of the argument as needed.
5. Write down as many questions and supplementary remarks as possible (kept separately from the preparation notes).
6. Prepare presentation notes (ideally, so that one can present even without any notes at hand).

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# Chapter 1

## Holomorphic Functions

### 1.1 Holomorphic functions of several variables

#### Notation and preliminaries

Let the  $n$ -dimensional complex space be

$$\mathbb{C}^n = \{ z = (z_1, \dots, z_n) \mid z_\nu \in \mathbb{C} \},$$

and write each coordinate as  $z_\nu = x_\nu + iy_\nu$  ( $x_\nu, y_\nu \in \mathbb{R}$ ,  $i = \sqrt{-1}$ ). Through the correspondence  $z \mapsto (x_1, y_1, \dots, x_n, y_n)$  taking real and imaginary parts, we identify  $\mathbb{C}^n \cong \mathbb{R}^{2n}$ .

Define the norm of  $z = (z_1, \dots, z_n)$  by

$$|z| = \left( \sum_{\nu=1}^n |z_\nu|^2 \right)^{1/2}.$$

For  $z, w \in \mathbb{C}^n$  the following hold:

$$|z_\nu| \leq |z| \leq \sum_{\nu=1}^n |z_\nu|, \quad (1.1)$$

$$|z + w| \leq |z| + |w|, \quad (1.2)$$

$$||z| - |w|| \leq |z - w|. \quad (1.3)$$

#### Holomorphic functions

**Def 1.1.** Let  $D \subseteq \mathbb{C}^n$ . We say that  $f(z) = f(z_1, \dots, z_n)$  is a *holomorphic function* on  $D$  if  $f$  is continuous on  $D$  and is holomorphic in each variable  $z_\nu$  separately (the other variables being held fixed).

#### Domains and polydiscs

**Def 1.2** (domain). A connected open subset of  $\mathbb{C}^n$  is called a *domain*.

*Note 1.1.* A topological space  $X$  is *connected* if every decomposition  $X = U \cup V$  with  $U \cap V = \emptyset$  and  $U, V$  open satisfies  $U = \emptyset$  or  $V = \emptyset$  (that is,  $X$  cannot be split into two nonempty open sets). For open subsets of  $\mathbb{C}^n$ , connectedness is equivalent to path-connectedness (any two points can be joined by a path within the set).

**Def 1.3** (polydisc). For  $a = (a_1, \dots, a_n) \in \mathbb{C}^n$  and  $r = (r_1, \dots, r_n)$  ( $r_\nu > 0$ ),

$$P(a, r) = \{ z \in \mathbb{C}^n \mid |z_\nu - a_\nu| < r_\nu, \nu = 1, \dots, n \}$$

is called the *polydisc* with center  $a$  and radius  $r$ .

*Note 1.2.* With the maximum norm  $\|z\| = \max_{1 \leq \nu \leq n} |z_\nu|$ , a polydisc is the “ball”  $\{ z \mid \|z - a\| < r \}$  with respect to  $\|z - a\|$  (when  $r_1 = \dots = r_n = r$ ).

## Wirtinger derivatives (complex derivatives) and the Cauchy–Riemann equations

For  $z_\nu = x_\nu + iy_\nu$ , define the *Wirtinger derivatives* by

$$\frac{\partial}{\partial z_\nu} = \frac{1}{2} \left( \frac{\partial}{\partial x_\nu} - i \frac{\partial}{\partial y_\nu} \right), \quad \frac{\partial}{\partial \bar{z}_\nu} = \frac{1}{2} \left( \frac{\partial}{\partial x_\nu} + i \frac{\partial}{\partial y_\nu} \right).$$

In what follows we split a  $C^1$  function  $f$  into real and imaginary parts,  $f = u + iv$  ( $u, v$  real-valued).

**Def 1.4** (holomorphy, Cauchy–Riemann equations). We say that  $f = u + iv$  is *holomorphic* in the variable  $z_\nu$  if

$$\frac{\partial f}{\partial \bar{z}_\nu} = 0.$$

This is equivalent to the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x_\nu} = \frac{\partial v}{\partial y_\nu}, \quad \frac{\partial u}{\partial y_\nu} = -\frac{\partial v}{\partial x_\nu}.$$

*Proof.* By definition,

$$\frac{\partial f}{\partial \bar{z}_\nu} = \frac{1}{2} \left( \frac{\partial}{\partial x_\nu} + i \frac{\partial}{\partial y_\nu} \right) (u + iv) = \frac{1}{2} \left[ \left( \frac{\partial u}{\partial x_\nu} - \frac{\partial v}{\partial y_\nu} \right) + i \left( \frac{\partial v}{\partial x_\nu} + \frac{\partial u}{\partial y_\nu} \right) \right].$$

A complex number is 0 exactly when its real and imaginary parts both vanish, so  $\partial f / \partial \bar{z}_\nu = 0$  is equivalent to the Cauchy–Riemann equations above.  $\square$

## Convergence of power series

*Note 1.3.* If, in a neighborhood of each point  $a$  of  $W$ ,  $f(z)$  can be expanded as a convergent power series

$$f(z) = \sum_{k_1, \dots, k_n \geq 0} c_{k_1 \dots k_n} (z_1 - a_1)^{k_1} \cdots (z_n - a_n)^{k_n},$$

then  $f(z)$  is a holomorphic function.

**Prop 1.1.** *If the power series  $\sum_{k_1, \dots, k_n \geq 0} c_{k_1 \dots k_n} (z_1 - a_1)^{k_1} \cdots (z_n - a_n)^{k_n}$  converges at some point  $w$ , then it converges absolutely for every  $z$  satisfying*

$$|z_\nu - a_\nu| < |w_\nu - a_\nu| \quad (\nu = 1, \dots, n).$$

*Proof.* We may assume  $a = 0$  (replace  $z_\nu - a_\nu$  by  $z_\nu$ ). Since the series converges at  $w$ , its terms are bounded: there is a constant  $C < \infty$  with

$$|c_{k_1 \dots k_n} w_1^{k_1} \cdots w_n^{k_n}| \leq C \quad (\forall k_1, \dots, k_n).$$

For  $|z_\nu| < |w_\nu|$ ,

$$|c_{k_1 \dots k_n} z_1^{k_1} \cdots z_n^{k_n}| = |c_{k_1 \dots k_n} w_1^{k_1} \cdots w_n^{k_n}| \prod_{\nu=1}^n \left| \frac{z_\nu}{w_\nu} \right|^{k_\nu} \leq C \prod_{\nu=1}^n \left| \frac{z_\nu}{w_\nu} \right|^{k_\nu},$$

and therefore

$$\sum_{k_1, \dots, k_n \geq 0} |c_{k_1 \dots k_n} z_1^{k_1} \cdots z_n^{k_n}| \leq C \prod_{\nu=1}^n \sum_{k=0}^{\infty} \left| \frac{z_\nu}{w_\nu} \right|^k = C \prod_{\nu=1}^n \frac{1}{1 - |z_\nu/w_\nu|} < \infty.$$

Hence the series converges absolutely.  $\square$

**Theorem 1: power-series expansion**

**Thm 1.** *Let  $W \subseteq \mathbb{C}^n$  be a domain. If  $f(z)$  is continuous on  $W$  and holomorphic in each variable  $z_\nu$  separately (the other variables being held fixed), then  $f(z)$  is a holomorphic function of the  $n$  variables  $z = (z_1, \dots, z_n)$ .*

*Proof.* Let  $a$  be an arbitrary point of  $W$ . Since  $W$  is open, we may choose  $r = (r_1, \dots, r_n)$  with

$$\{z \mid |z_\nu - a_\nu| \leq r_\nu, \nu = 1, \dots, n\} \subset W.$$

As  $f$  is holomorphic in  $z_1$ , the Cauchy integral formula in the  $z_1$ -plane gives

$$f(z_1, z_2, \dots, z_n) = \frac{1}{2\pi i} \oint_{|w_1 - a_1| = r_1} \frac{f(w_1, z_2, \dots, z_n)}{w_1 - z_1} dw_1.$$

Repeating this for each variable, and using that  $f$  is continuous so that the iterated integral equals the multiple integral (the order of integration may be exchanged), we obtain

$$f(z) = \frac{1}{(2\pi i)^n} \oint_{|w_1 - a_1| = r_1} \cdots \oint_{|w_n - a_n| = r_n} \frac{f(w_1, \dots, w_n)}{(w_1 - z_1) \cdots (w_n - z_n)} dw_1 \cdots dw_n.$$

Now, noting that  $|z_\nu - a_\nu| < r_\nu = |w_\nu - a_\nu|$ , each factor expands as

$$\frac{1}{w_\nu - z_\nu} = \frac{1}{w_\nu - a_\nu} \cdot \frac{1}{1 - \frac{z_\nu - a_\nu}{w_\nu - a_\nu}} = \sum_{k=0}^{\infty} \frac{(z_\nu - a_\nu)^k}{(w_\nu - a_\nu)^{k+1}}.$$

Substituting these and integrating term by term,

$$f(z) = \sum_{k_1, \dots, k_n \geq 0} c_{k_1 \dots k_n} (z_1 - a_1)^{k_1} \cdots (z_n - a_n)^{k_n},$$

where

$$c_{k_1 \dots k_n} = \frac{1}{(2\pi i)^n} \oint_{|w_1 - a_1| = r_1} \cdots \oint_{|w_n - a_n| = r_n} \frac{f(w_1, \dots, w_n)}{(w_1 - a_1)^{k_1+1} \cdots (w_n - a_n)^{k_n+1}} dw_1 \cdots dw_n.$$

Hence  $f$  is expanded as a convergent power series in a neighborhood of each point of  $W$ , and is therefore a holomorphic function of  $n$  variables.  $\square$